(which is meaningless), (ii) the shift-shielding relation (which allows us to transform chemical shieldings into shifts) and (iii) the overall width of the lineshapes. None of these parameters are essential to the interpretation and assignment of the peaks in the experimental spectra (although they facilitate the plotting of the computational chemical shielding data on the experimental chemical shift axis). One could also choose to merely translate the chemical shielding values to obtain the chemical shift (in notation of the manuscript, this means leaving the parameters a fixed at 1.0 and choosing b such that, for example, the location of the methyl or methoxy peak matches exactly the experimental value). Another option would be to fit the shift-shielding relation to external references, as is often done. We chose to fit the parameters to the non-quantitative experimental spectrum regardless, because if one were able to measure a quantitative experimental NMR spectrum, then the fitting procedure in its current form would be optimal (because in that case, it is expected that the method would produce better results as the experimental spectrum is also quantitative). If one wishes to use this method but change it to also fit the individual amplitudes of the lineshapes used to the non-quantitative experimental spectrum, than this may easily be done by slightly adapting the provided Python fitting script by giving the MSE minimization procedure the additional parameters to be fit (which are the individual lineshape amplitudes A_n).

S2 Internal validation: dynamic vs. static model within g_S approach

We compare the model spectra obtained using static and dynamic methods to produce carbon spectra with the g_S approach. The dynamic method considers 21 snapshots spaced 10 ps selected from a 0.2 ns long force field MD run (0.5 fs timestep), as the dynamic carbon chemical shieldings were validated to be converged to within 2 ppm from that point on. This was deemed to be sufficient for our purposes given the range of chemical shifts that the carbon resonances occur in for this system. The shieldings are then time-averaged for each